

K.J. College Of Engineering And Management Research ,Pune.

Department of Engineering Sciences.

DESIGN THINKING AND IDEA LAB PROJECT

Project Guide : Dr.Sangita Thakre

Project By :

- 1. Affan Shaikh**
- 2. Yash Aher**
- 3. Swaraj Akhade**
- 4. Aryaan Shaikh**
- 5. Aditya Balsaraf**
- 6. Rushikesh Bandgar**
- 7. Kabir Bhoite**



TRINITY

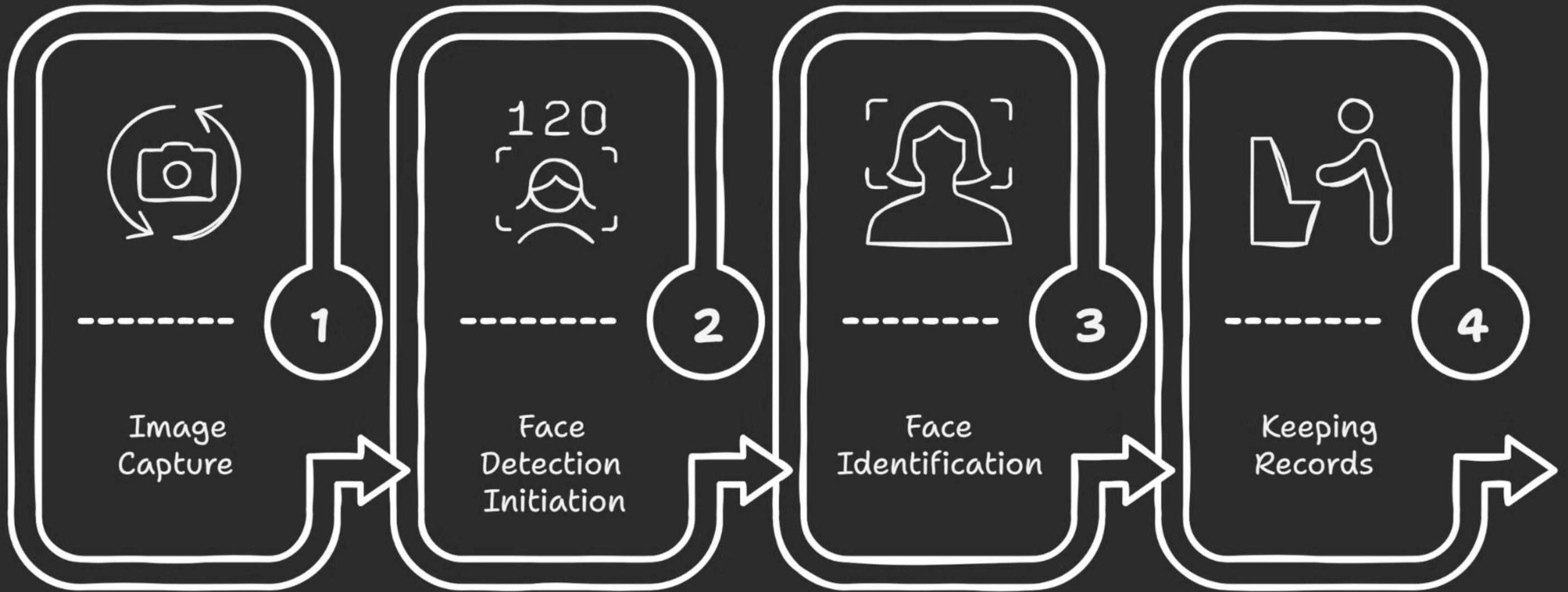
FaceLog

Facial Recognition-Based Student Attendance System.

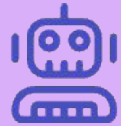
Problem Statement :

Manual attendance marking systems are time-consuming, error-prone, and inefficient, especially for large groups. Issues like proxy attendance, data mismanagement, and human errors make tracking unreliable.

System Overview



Objectives:



Automation

- Eliminate manual recording of attendance.
- To develop an automated attendance system using face recognition.



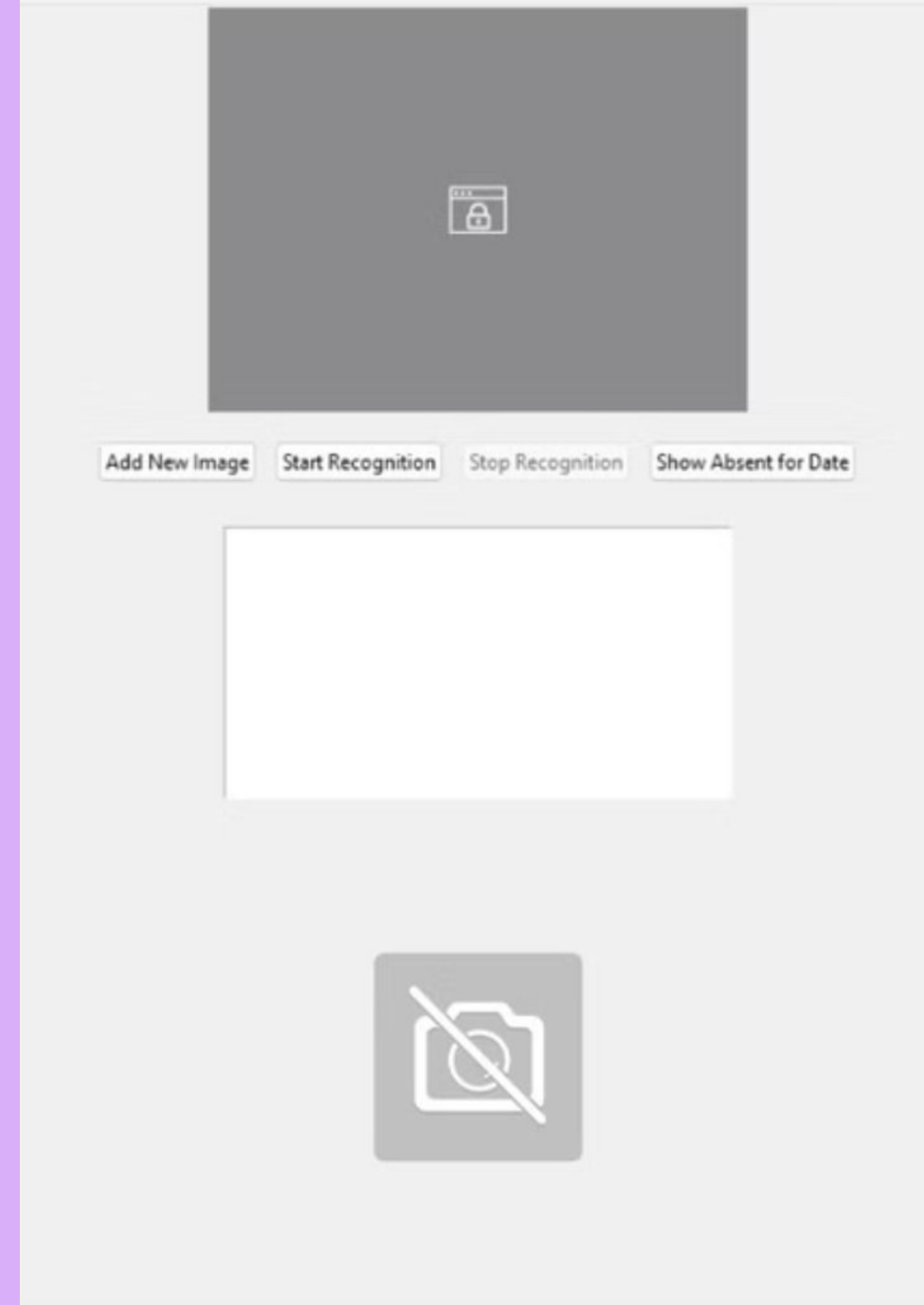
Accuracy

- Prevent proxy attendance, reduce human errors.
- To provide real-time attendance tracking and reporting



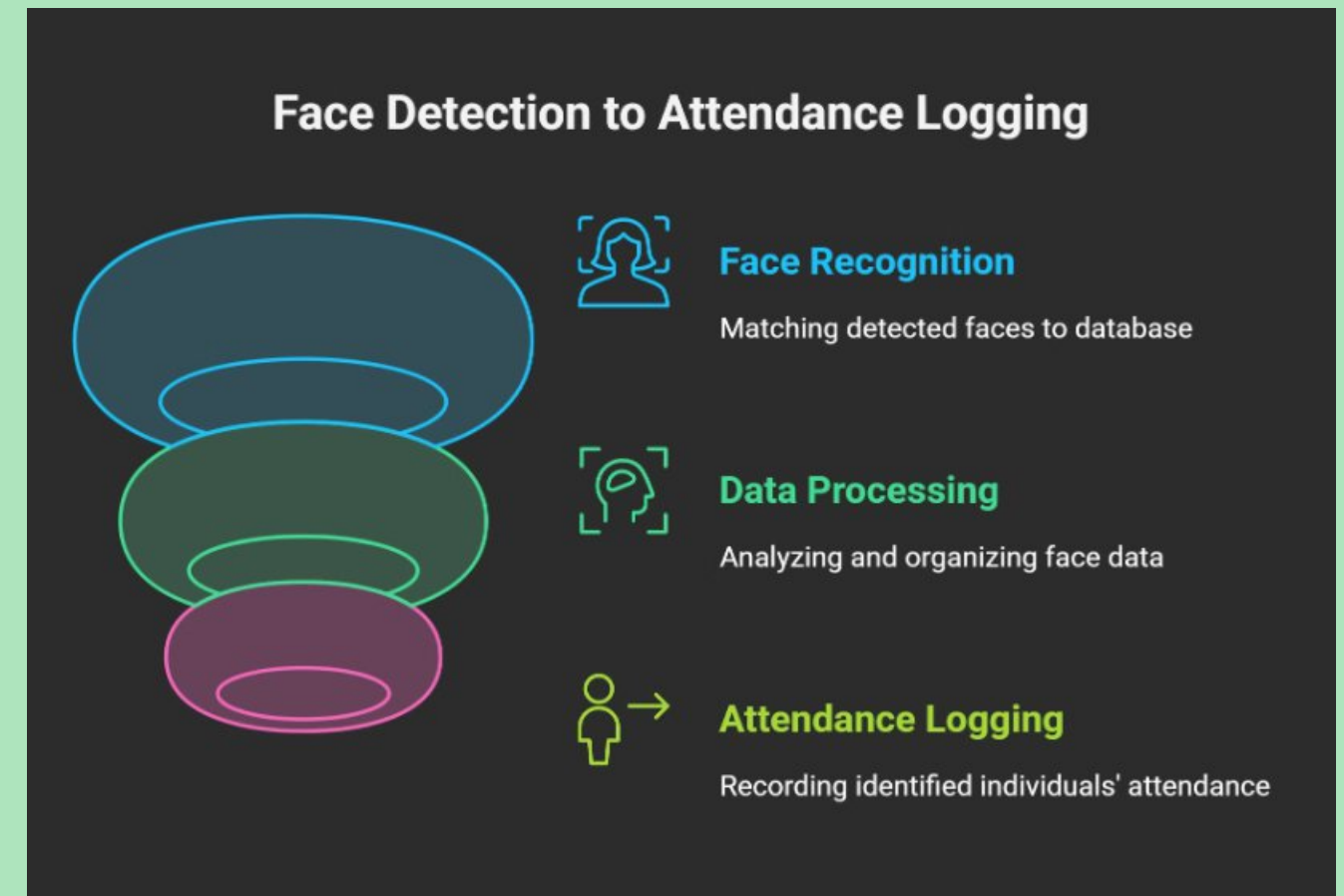
Efficiency

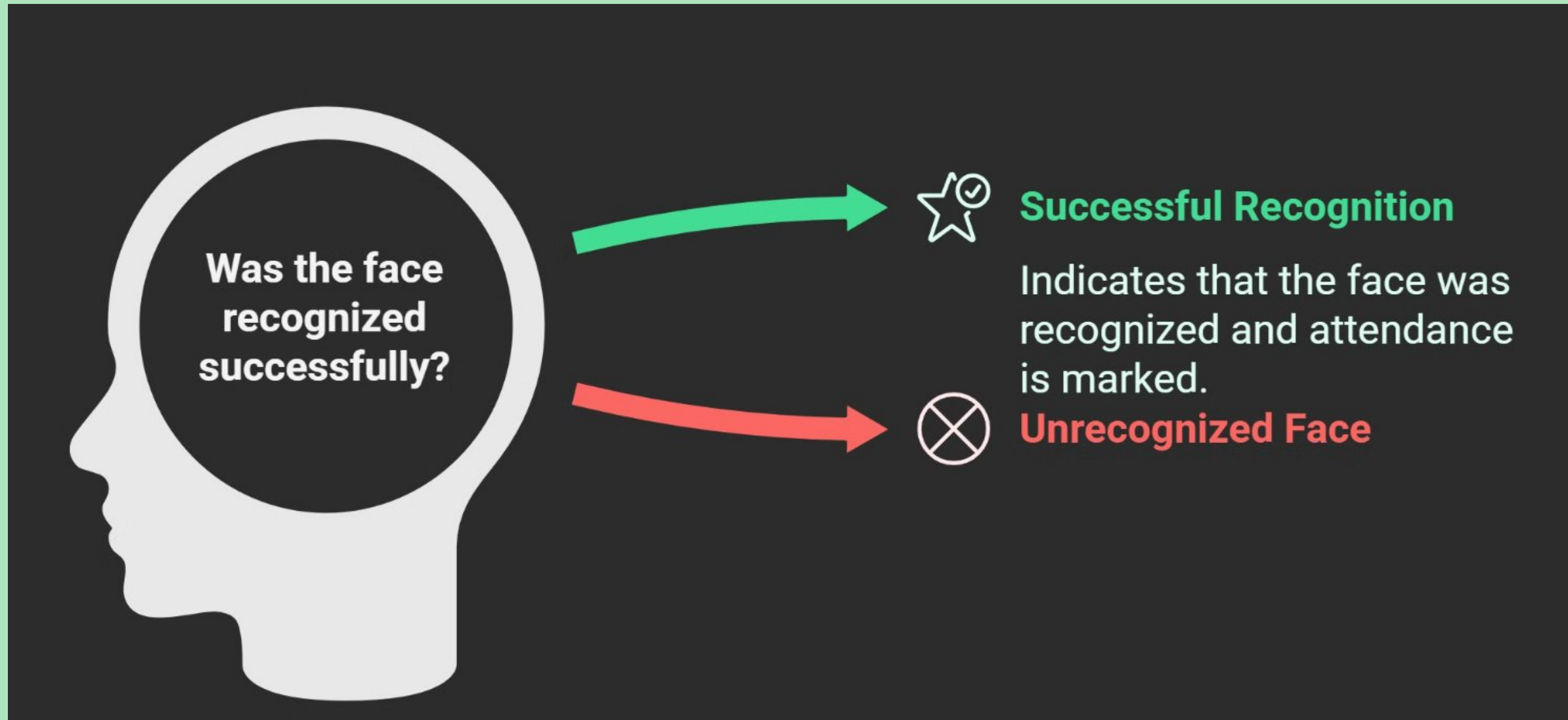
- Reduce time for attendance marking.
- Make it Easy to use and scalable.



System Capabilities

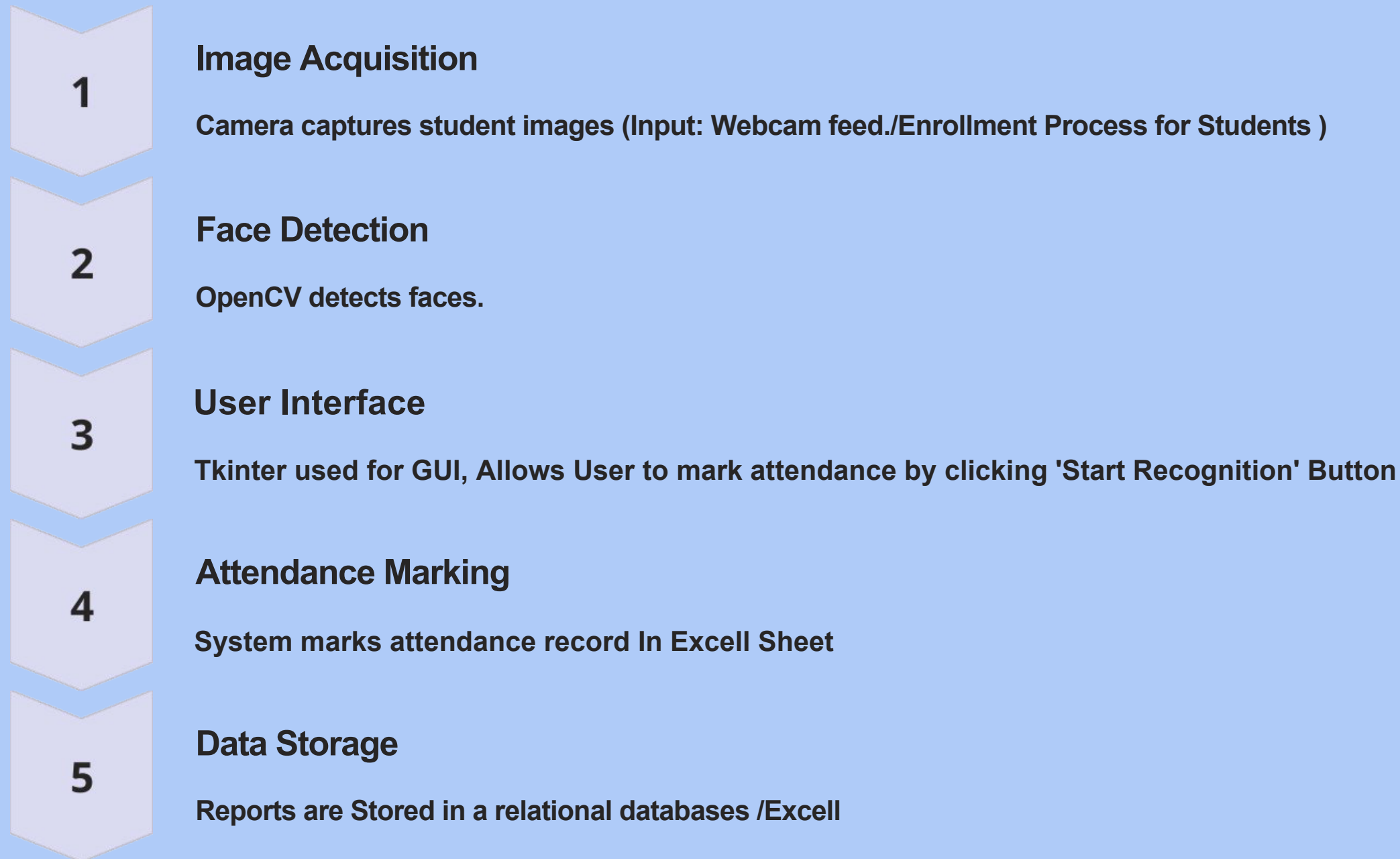
- Capture images of students and store them in the database.
- Use face recognition to identify students in real-time.
- Export attendance data to CSV.
- Mark attendance and generate reports.





The user interface displays attendance records. The server processes images and if not matched with Database No attendance is marked.

System Workflow





Expected Outcomes

Accurate Records

Eliminates proxy attendance and errors.

Reduced Time

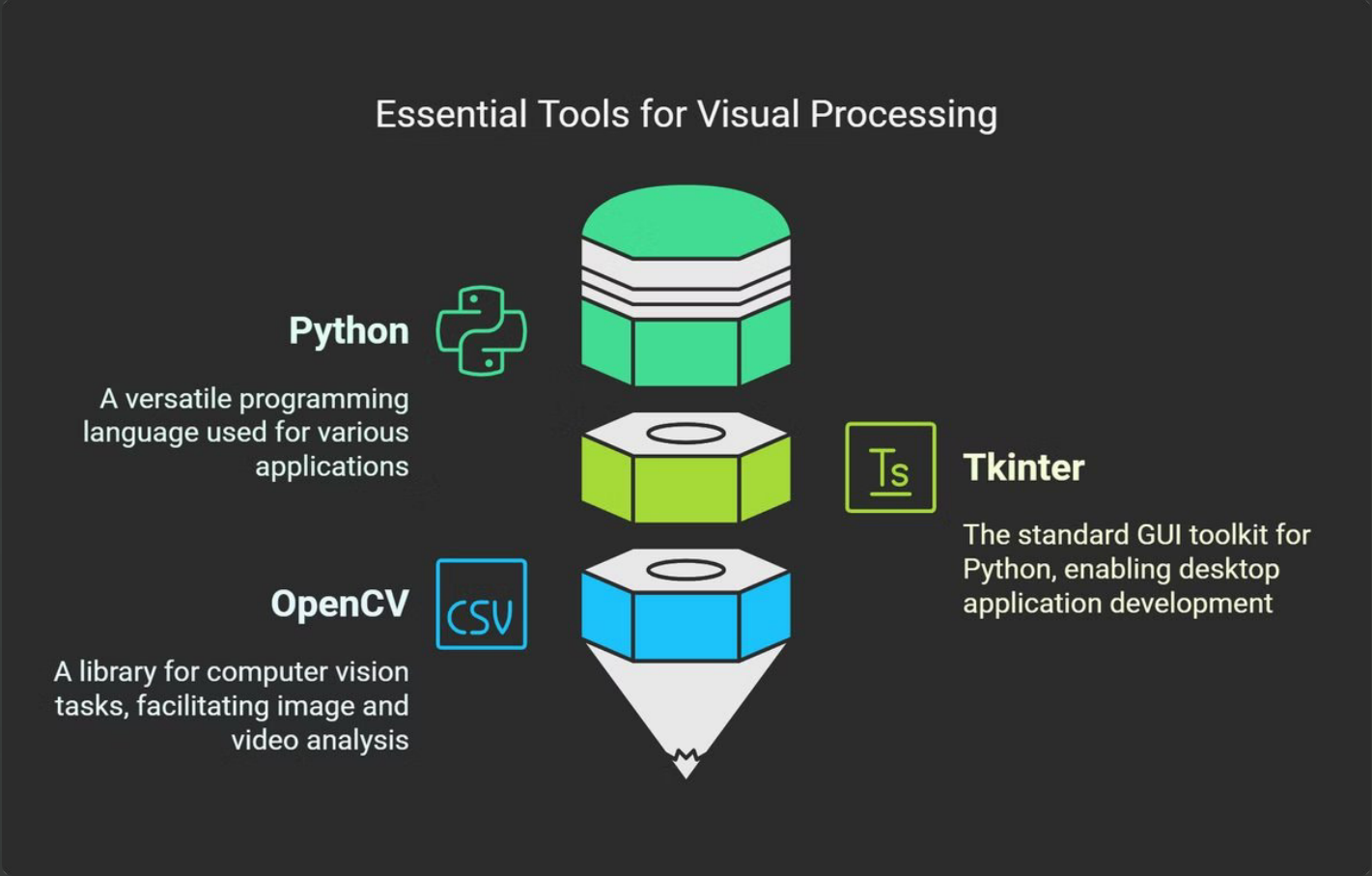
Significantly reduces attendance time.

Improved Efficiency

Streamlines process for students/instructors.

Components Used :

Programming Language	Python
Libraries/Frameworks	OpenCV, Face Recognition, Pandas, Tkinter
Tools	Visual Studio Code, Git
Hardware	Camera, Processing Unit



Challenges and Future Enhancements:

Challenges:

- Handling lighting and pose.
- Managing large datasets.
- Ensuring security/privacy.
- Managing lag in live video feed.

Future Enhancements:

- Integration with cloud storage for data backup
- Implementing emotion recognition.
- Developing a mobile app

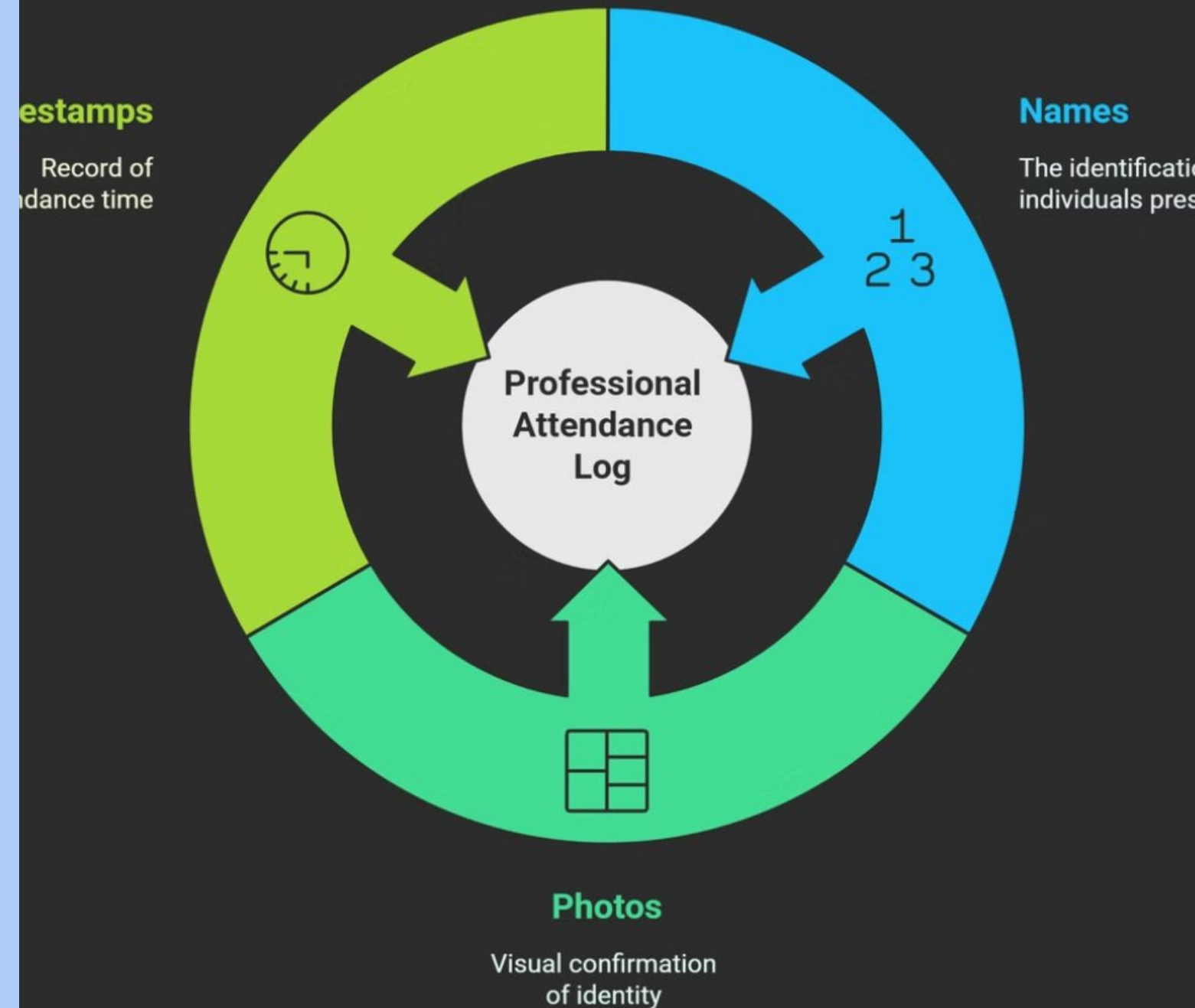
Conclusion

Our system provides an efficient, accurate, automated solution.

It addresses limitations, saving time and reducing errors. Real-time monitoring and data analysis offers insights.

Future enhancements will improve its capabilities.

Components of a Professional Attendance Log



References

Python: <https://www.python.org/doc/>

OpenCV: <https://docs.opencv.org/>

Face Recognition Library: https://github.com/ageitgey/face_recognition

Tkinter (GUI Framework): <https://github.com/joeVenner/FaceRecognition-GUI-APP.git>

Pandas (Data Handling): <https://pandas.pydata.org/docs/>

THANK YOU!